Characteristics of objectively-measured naturalistic light exposure patterns in U.S. adults: A cross-sectional analysis of two cohorts

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Abstract

Light is an environmental feature important for human physiology. Investigation of how light affects population health requires exposure assessment and personal biomonitoring efforts. Here, we derived measures of amount, duration, regularity, and timing from objective personal light (lux) measurement in >4000 participants across two United States (US)-based cohort studies, the Multi-Ethnic Study of Atherosclerosis (MESA) and the Hispanic Community Health Study / Study of Latinos (HCHS/SOL), encompassing eight geographic regions. Objective light and actigraphy data were collected over a week using wrist-worn devices (Actiwatch Spectrum). Cohort-stratified light exposure metrics were analyzed in relation to sex, season, time-of-day, location, and demographic and sleep health characteristics using Spearman correlation and linear and logistic regressions (separately by cohort) adjusted for age, sex (where applicable), and exam site. Light exposure showed sex-specific patterns and had seasonal, diurnal, geographic, and demographic and sleep health-related correlates. Results between independent cohorts were strongly consistent, supporting the utility and feasibility of light biomonitoring. These findings provide a fundamental first characterization of light exposure patterns in a large US sample and will inform future work to incorporate light as a biologically relevant exposure in environmental public health and key component of the human exposome.